

Week 7
Dynamics

ENN581 Robot Motion, Control and Planning

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School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

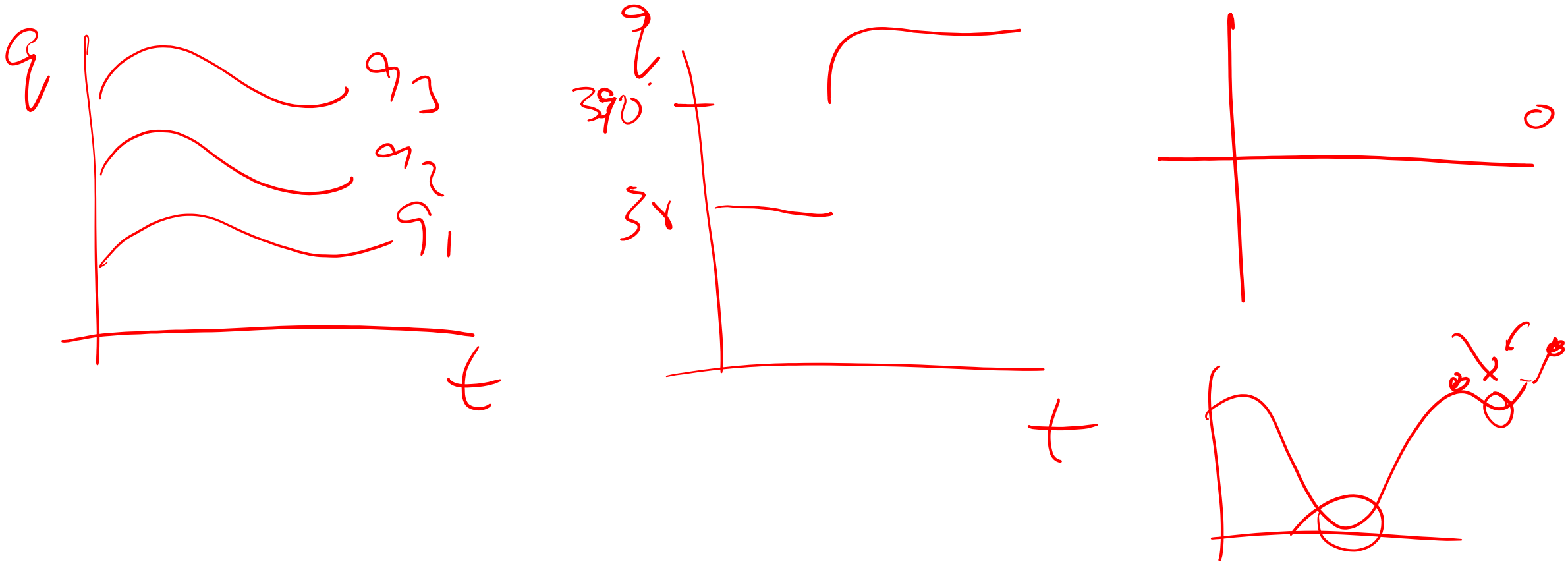
Mid-semester Exam this afternoon

Venue: GP-S-620

Time: 4:00 pm

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How was assessment 2.3?



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Robotics Toolbox for Python



Robotics without the cruft

A high-productivity framework for robotics research and education.

TRY IT NOW JUPYTERLITE | PYPI V1.4.2 | DOCS VIEW ONLINE

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THIS RELEASE

1.4.2
Aug 26, 2026 | 22 files

1.4.1
Aug 25, 2026 | 22 files

1.4.0
Aug 23, 2026 | 24 files

1.3.1
Jul 3, 2026 | 17 files

1.3.0
Jun 14, 2026 | 17 files

1.1.1
May 11, 2024 | 2 files

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Dr. KM Digumarti

A recap of what we learnt so far

Forward Kinematics: Given the positions of the joints, where is the end effector?

$$^{0}\mathbf{T}_{n} \text{ or } \boldsymbol{\xi}_{n} = \mathcal{K}(\mathbf{q}),\ \mathbf{q} = \{q_i, i \in [1, \cdots, n]\}$$

Inverse Kinematics: Given the tool pose, what are the joint positions to get there?

$$\mathbf{q} = \mathcal{K}^{-1}(\boldsymbol{\xi}_{n})$$

Trajectory generation: Along what path should a manipulator move?

$$\mathbf{q}(t) \text{ and } \boldsymbol{\xi}_{n}(t)$$

Velocity Kinematics: How fast is the robot moving (FK) or should it go (IK)?

$$^{0}\boldsymbol{\nu} = {}^{0}\mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

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Today's lecture

Dynamics

How fast should the joints accelerate? (Forward)

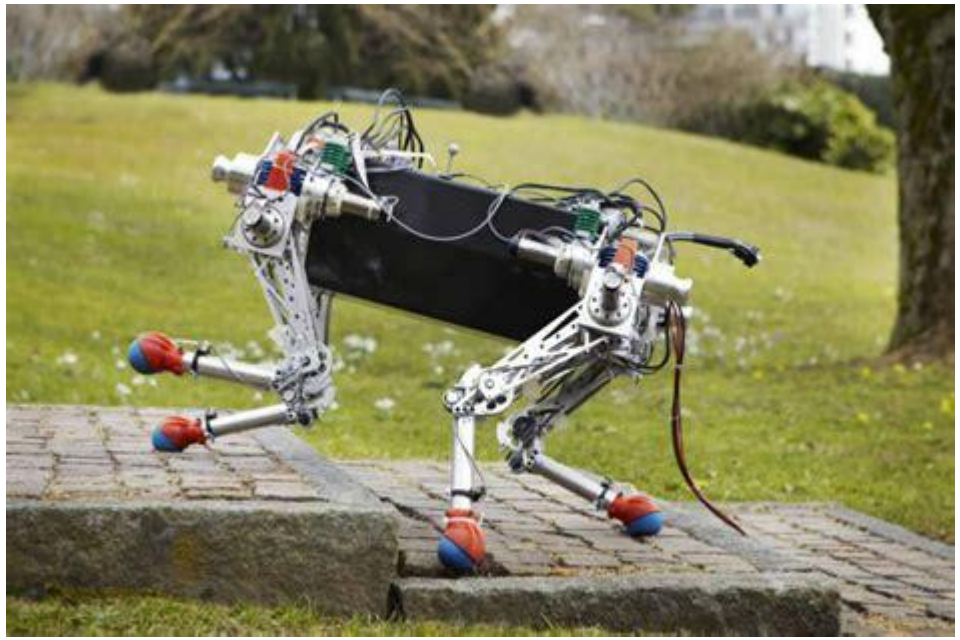
How much torque should the motors apply? (Inverse)

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Why study dynamics?







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Forward Dynamics

- Given: joint forces & torques $\mathbf{Q}$.
- Aim: to determine the motion of the manipulator $\mathbf{q}$, $\dot{\mathbf{q}}$, $\ddot{\mathbf{q}}$.

- Useful for simulation.

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Inverse dynamics

- Given: joint positions $\mathbf{q}$, joint velocities $\dot{\mathbf{q}}$, and joint accelerations $\ddot{\mathbf{q}}$.
- Aim: to determine the joint torques $\mathbf{Q}$.

- Useful in the control of robots.

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Joint torques in dynamics

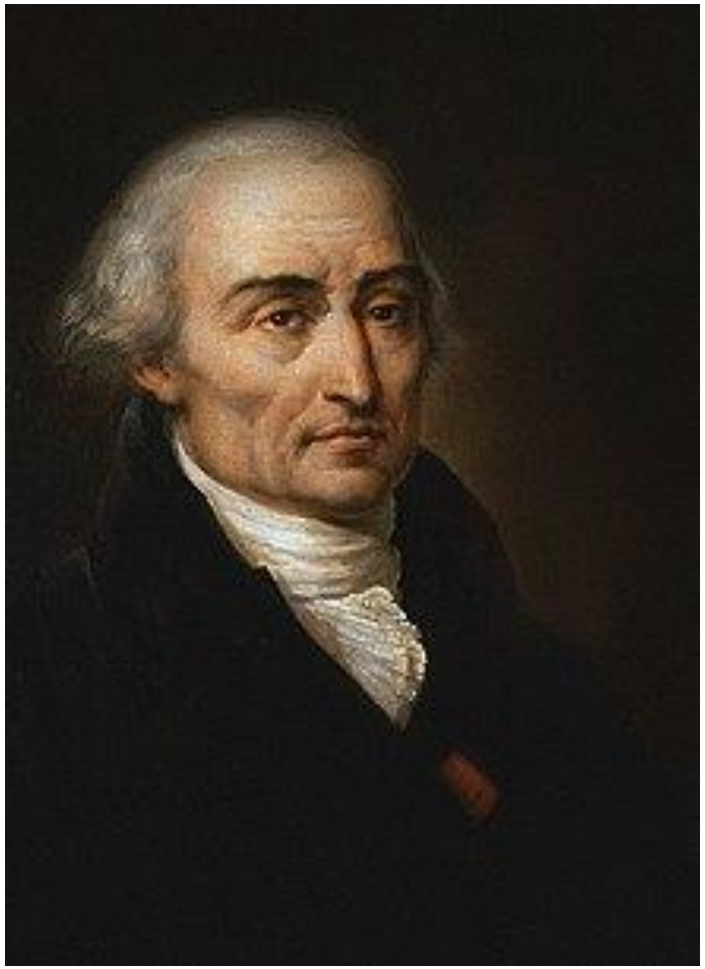
Joint actuators are balancing torques from many sources:

1. Dynamic torques – those that arise due to motion
   - Inertial (proportional to acceleration)
   - Centripetal (constrain body to rotate about a point, directed towards the centre)
   - Coriolis (result of interaction between two rotating links)
2. Static torques – arising from an external load
3. Gravity torques – due to gravity
4. Frictional torques – due to friction at joints
5. Other external forces/torques – arising from the task

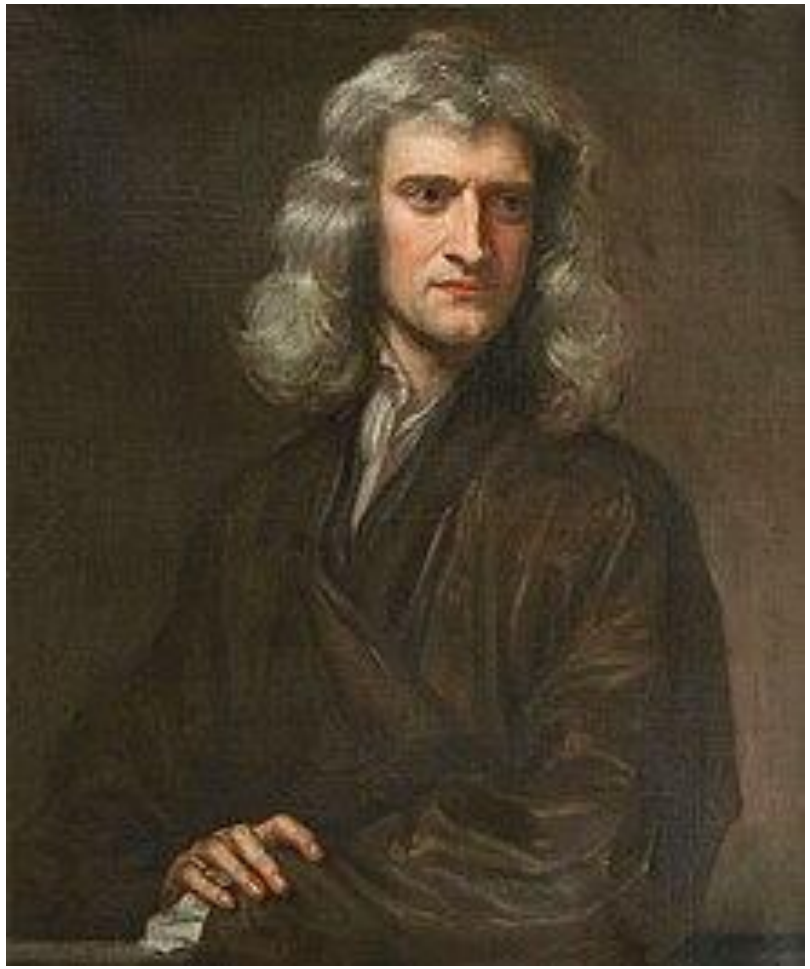
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Inverse Dynamics

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Joseph Louis Lagrange



Isaac Newton



Leonhard Euler

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Approaches to studying dynamics

Lagrangian formulation

Energy based approach

- Multi-body robot is seen as a whole
- Internal reaction forces between links do not show up
- Closed form equations are directly obtained
- Best suited for study of dynamic properties and analysis of control

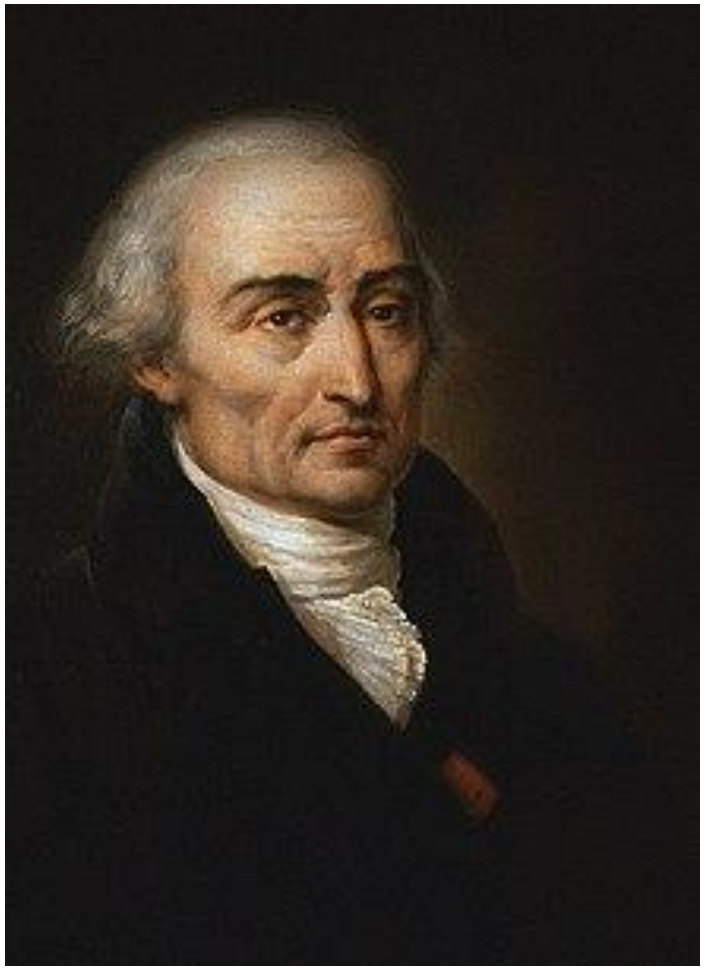
Newton-Euler formulation

Based on balance of forces and torques

- Equations are written for each link
- Enables inverse dynamics in real time
- Iterative numerical approach
- Best suited for implementing model-based control

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Lagrangian formulation for dynamics



Joseph Louis Lagrange

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The key quantity in Lagrangian dynamics

- Think in terms of work done by the system and energy stored in the system.

- The Lagrangian is a scalar function defined as follows

$$L(\mathbf{q}, \dot{\mathbf{q}}) \coloneqq K(\mathbf{q}, \dot{\mathbf{q}}) - P(\mathbf{q})$$

Kinetic energy

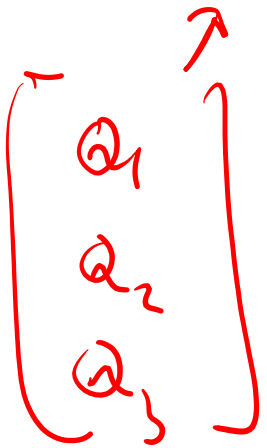
Potential energy

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The Lagrangian equations of motion

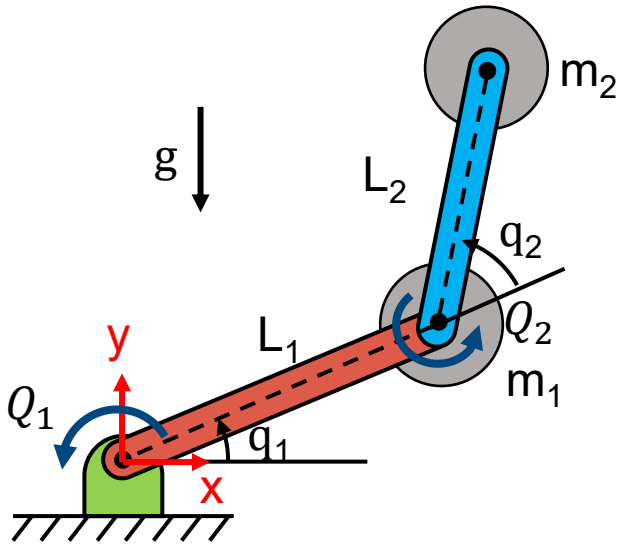
The vector of joint forces and torques, expressed in terms of the Lagrangian is

$$\mathbf{Q} = \frac{d}{dt}\left(\frac{\partial L(\mathbf{q}, \dot{\mathbf{q}})}{\partial \dot{\mathbf{q}}}\right) - \frac{\partial L(\mathbf{q}, \dot{\mathbf{q}})}{\partial \mathbf{q}}$$



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Example: Planar two-link manipulator



Assumptions:

- All the mass of the robot is concentrated in point-masses m_1 and m_2
- Gravity is pointing towards -y

Given:

- $(q_1, \dot{q}_1, \ddot{q}_1)$ and $(q_2, \dot{q}_2, \ddot{q}_2)$

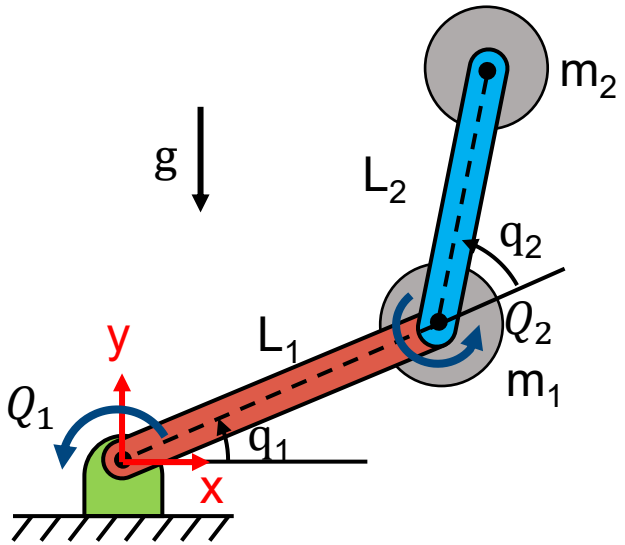
Aim:

- Determine torques Q_1, Q_2

To formulate the Lagrangian, we begin by writing the kinetic and potential energy of the two point-masses.

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Example: Planar two-link manipulator



Assume that m_1 and m_2 are point masses

The position of point-mass m_1 is

$$x_1 = L_1 \cos(q_1)$$

$$y_1 = L_1 \sin(q_1)$$

Its velocity is

$$\dot{x}_1 = -L_1 \sin(q_1)\,\dot{q}_1$$

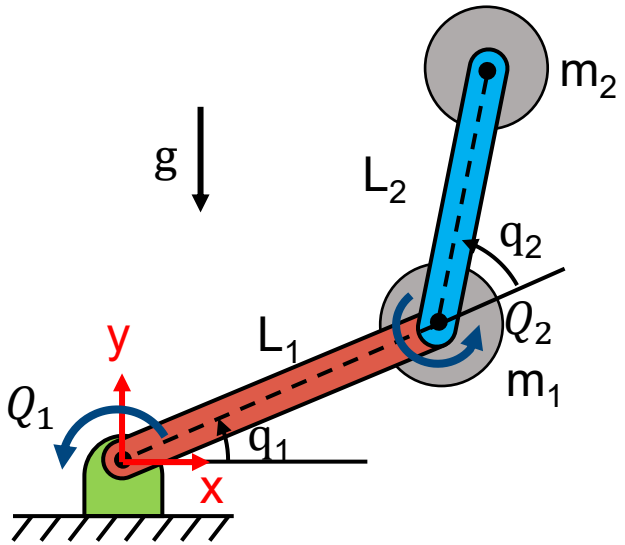
$$\dot{y}_1 = L_1 \cos(q_1)\,\dot{q}_1$$

Therefore, its kinetic energy can be written as

$$K_1 = \frac{1}{2} m_1 \left(\dot{x}_1^{\,2} + \dot{y}_1^{\,2}\right) = \frac{1}{2} m_1 L_1^{\,2} \dot{q}_1^{\,2}$$

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Example: Planar two-link manipulator



Assume that m_1 and m_2 are point masses

The position of point-mass m_2 is

$$x_2 = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$

$$y_2 = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

Its velocity is

$$\dot{x}_2 = -L_1 \sin(q_1)\dot{q}_1 - L_2 \sin(q_1 + q_2)(\dot{q}_1 + \dot{q}_2)$$

$$\dot{y}_2 = L_1 \cos(q_1)\dot{q}_1 + L_2 \cos(q_1 + q_2)\,(\dot{q}_1 + \dot{q}_2)$$

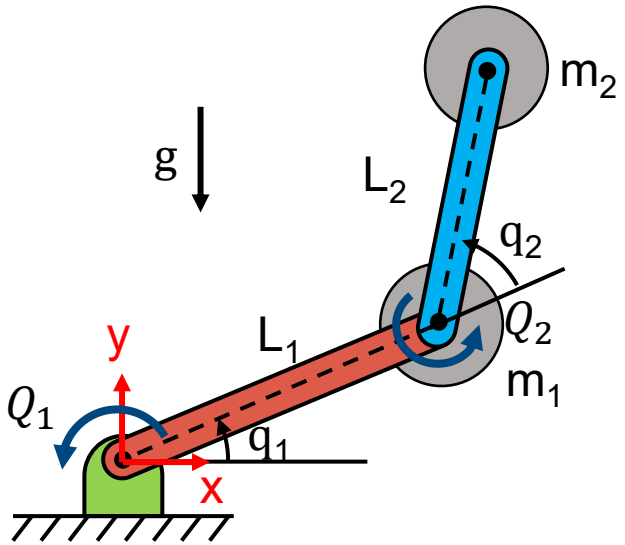
Therefore, its kinetic energy can be written as

$$K_2 = \frac{1}{2} m_2 \left(\dot{x}_2{}^2 + \dot{y}_2{}^2\right)$$

$$= \frac{1}{2} m_2 \left(\left(L_1{}^2 + 2L_1 L_2 \cos q_2 + L_2{}^2\right)\dot{q}_1{}^2 + 2\left(L_2{}^2 + L_1 L_2 \cos q_2\right)\dot{q}_1 \dot{q}_2 + L_2{}^2 \dot{q}_2{}^2\right)$$

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Example: Planar two-link manipulator



Assume that m_1 and m_2 are point masses

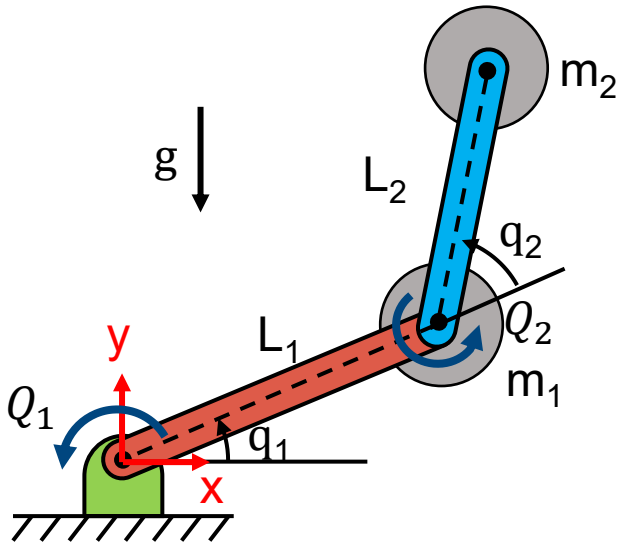
The potential energies are

$$P_1 = m_1 g y_1 = m_1 g L_1 \sin(q_1)$$

$$P_2 = m_2 g y_2 = m_2 g (L_1 \sin(q_1) + L_2 \sin(q_1 + q_2))$$

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Example: Planar two-link manipulator



Assume that m_1 and m_2 are point masses

The Lagrangian is $L = K - P$

$$L(\mathrm{q}, \dot{\mathrm{q}}) = \sum_{i=1}^{2} (K_i - P_i)$$

$$= K_1 - P_1 + K_2 - P_2$$

$$L(\mathrm{q}, \dot{\mathrm{q}})$$

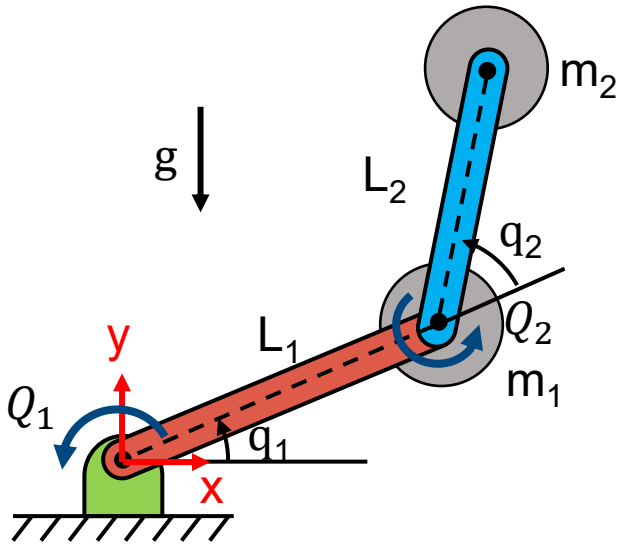
$$= \left(\frac{1}{2} m_1 L_1^{\,2} \dot{q}_1^{\,2} - m_1 g L_1 \sin(q_1)\right)$$

$$+ \left(\frac{1}{2} m_2 \left(\left(L_1^{\,2} + 2L_1 L_2 \cos q_2 + L_2^{\,2}\right)\dot{q}_1^{\,2} + 2\left(L_2^{\,2} + L_1 L_2 \cos q_2\right)\dot{q}_1 \dot{q}_2 + L_2^{\,2} \dot{q}_2^{\,2}\right)\right.$$

$$\left. - m_2 g (L_1 \sin(q_1) + L_2 \sin(q_1 + q_2))\right)$$

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Example: Planar two-link manipulator



Assume that m_1 and m_2 are point masses

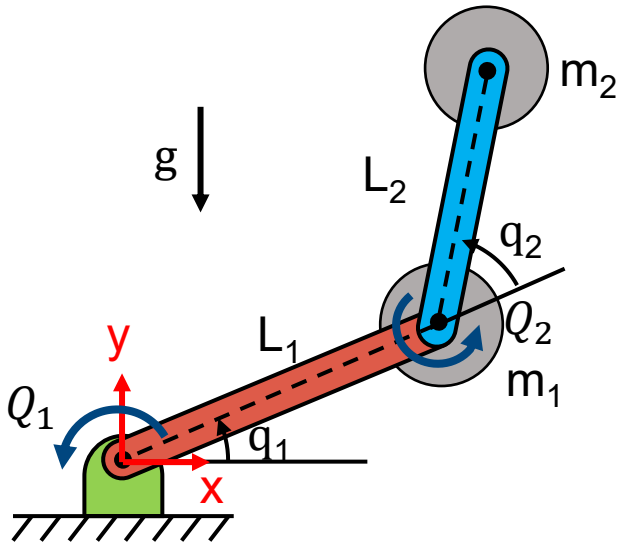
The equations of motion are given as

$$Q_1 = \frac{d}{dt}\left(\frac{\partial L(\mathrm{q},\dot{\mathrm{q}})}{\partial \dot{\mathrm{q}}_1}\right) - \frac{\partial L(\mathrm{q},\dot{\mathrm{q}})}{\partial \mathrm{q}_1}$$

$$Q_2 = \frac{d}{dt}\left(\frac{\partial L(\mathrm{q},\dot{\mathrm{q}})}{\partial \dot{\mathrm{q}}_2}\right) - \frac{\partial L(\mathrm{q},\dot{\mathrm{q}})}{\partial \mathrm{q}_2}$$

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Example: Planar two-link manipulator



Assume that m_1 and m_2 are point masses

The equations of motion are given as

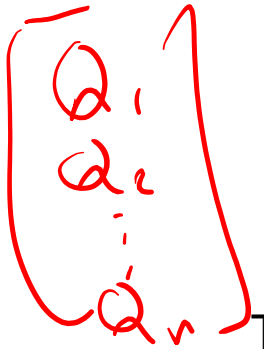
$$Q_1 = \left(m_1 L_1^2 + m_2\left(L_1^2 + 2L_1 L_2 \cos q_2 + L_2^2\right)\right)\ddot{q}_1 + m_2\left(L_1 L_2 \cos q_2 + L_2^2\right)\ddot{q}_2 - m_2 L_1 L_2 \sin q_2 \left(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2\right) + (m_1 + m_2) L_1 g \cos q_1 + m_2 g L_2 \cos(q_1 + q_2)$$

$$Q_2 = m_2\left(L_1 L_2 \cos q_2 + L_2^2\right)\ddot{q}_1 + m_2 L_2^2 \ddot{q}_2 + m_2 L_1 L_2 \dot{q}_1^2 \sin q_2 + m_2 g L_2 \cos(q_1 + q_2)$$

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General equations of motion





$$\underset{n\times 1}{\boldsymbol{Q}} = \underset{n\times n}{\mathbf{M}(\mathbf{q})}\underset{n\times 1}{\ddot{\mathbf{q}}} + \underset{n\times 1}{\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})} + \underset{n\times 1}{\boldsymbol{g}(\mathbf{q})}$$

Joint Torques	Mass matrix	Velocity product	Gravity term

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} m_1 L_1{}^2 + m_2\left(L_1{}^2 + 2L_1 L_2 \cos q_2 + L_2{}^2\right) & m_2\left(L_1 L_2 \cos q_2 + L_2{}^2\right) \\ m_2\left(L_1 L_2 \cos q_2 + L_2{}^2\right) & m_2 L_2{}^2 \end{bmatrix}$$

$$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) = \begin{bmatrix} -m_2 L_1 L_2 \sin q_2 \left(2\dot{q}_1\dot{q}_2 + \dot{q}_2{}^2\right) \\ m_2 L_1 L_2 \dot{q}_1{}^2 \sin q_2 \end{bmatrix}$$

$$\boldsymbol{g}(\mathbf{q}) = \begin{bmatrix} (m_1 + m_2) L_1 g \cos q_1 + m_2 g L_2 \cos(q_1 + q_2) \\ m_2 g L_2 \cos(q_1 + q_2) \end{bmatrix}$$

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General equations of motion

$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q}) + \boldsymbol{f}(\dot{\mathbf{q}}) + \mathbf{J}^{\mathrm{T}}(\mathbf{q})\boldsymbol{w}$$

Mass matrix

Velocity product

Gravity term

Friction

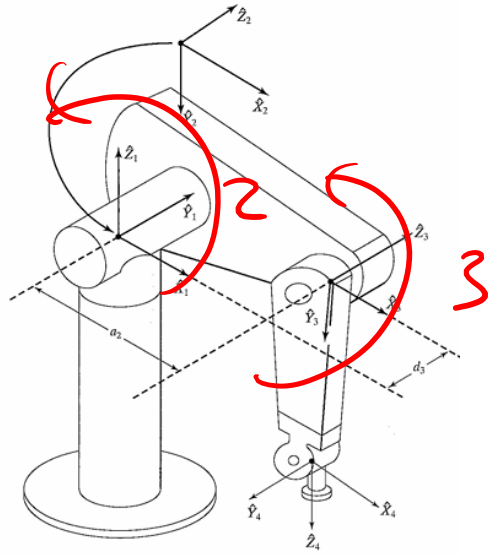
External forces (e.g., from contacts)

Other contributors to potential energy also appear here. E.g., spring forces in joints

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Gravity term $g(\mathbf{q})$

- Usually the dominant quantity
- Present even when the robot is stationary



Rest position

```
array([      0,     31.64,     6.035,         0,   0.02825,         0])
```

Arm stretched out

```
array([      0,     46.01,     8.772,         0,   0.02825,         0])
```

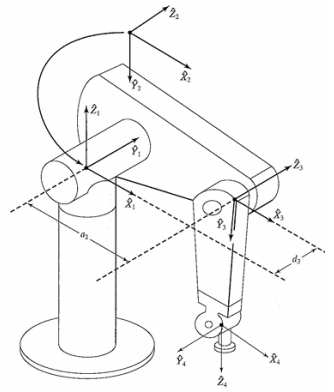
Arm pointing up

```
array([      0,   -0.7752,    0.2489,         0,         0,         0])
```

shoulder elbow

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Mass matrix M(q)



- Joint-space inertia matrix
- Symmetric and positive definite

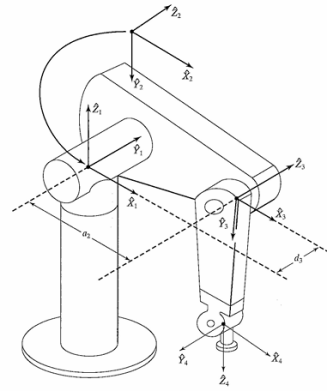
```
array([[   3.659,   -0.4044,    0.1006, -0.002517,        0,        0],
       [ -0.4044,    4.414,    0.3509,        0,  0.00236,        0],
       [  0.1006,   0.3509,    0.9378,        0,  0.00148,        0],
       [-0.002517,        0,         0,   0.1925,        0, 2.828e-05],
       [       0,  0.00236,   0.00148,        0,   0.1713,        0],
       [       0,        0,         0, 2.828e-05,        0,   0.1941]])
```

The diagonal terms describe inertia experienced by joint j

The off-diagonal terms are the products of inertia - couple the acceleration of joint j to force on joint i

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Friction $f(\dot{\mathbf{q}})$



Friction on the first joint when it is moving at 1 rad/s from rest configuration

```
array([  -30.53,     0,     0,     0,     0,     0])
```

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Centripetal and Coriolis terms

Velocity coupling

$$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) = \begin{bmatrix} -m_2 L_1 L_2 \sin q_2 \, \dot{q}_2^{\,2} - 2 m_2 L_1 L_2 \sin q_2 \, \dot{q}_1 \dot{q}_2 \\ m_2 L_1 L_2 \dot{q}_1^{\,2} \sin q_2 \end{bmatrix} = C(\mathbf{q})[\dot{\mathbf{q}}^2] + B(\mathbf{q})[\dot{\mathbf{q}}\dot{\mathbf{q}}]$$

$n \times 1$

$n+n$, $n+1$, $n+n(n-1)/2$, $n(n-1)/2+1$

$\frac{n(n-1)}{2} \times 1$

- Centripetal terms have squares of velocities.
- Coriolis terms have products of velocities.

$n = 3$

$\begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \dot{q}_n^2 \end{bmatrix}$ $\begin{bmatrix} \dot{q}_1\dot{q}_2 \\ \dot{q}_1\dot{q}_3 \\ \dot{q}_1\dot{q}_3 \end{bmatrix}$ ${}^nC_2 = \frac{n \times (n-1)}{2}$

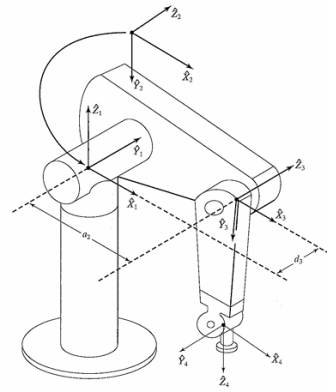
for 2D example C:

$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Refer to section 8.1 of Lynch and Park for more details

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Centripetal and Coriolis terms



Velocity coupling

C matrix in PUMA 560 when it is moving at 1 rad/s from rest configuration

```
array([[  0.3607, -0.09566, -0.09566, 0.0005445, -0.0004321,  2e-05],
       [       0,   0.3858,   0.3858,         0, -4.134e-05,      0],
       [       0,        0,        0,         0, -0.0009207,      0],
       [-3.584e-05,      0,        0,         0,          0, -1.414e-05],
       [       0, 0.0009207, 0.0009207,        0,          0,      0],
       [   2e-05,        0,        0, 1.414e-05,          0,      0]])
```

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Manipulator dynamics in task (cartesian) space

The equation of motion is described in the configuration (joint) space as

$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q})$$

It may be desirable to express them in the task space

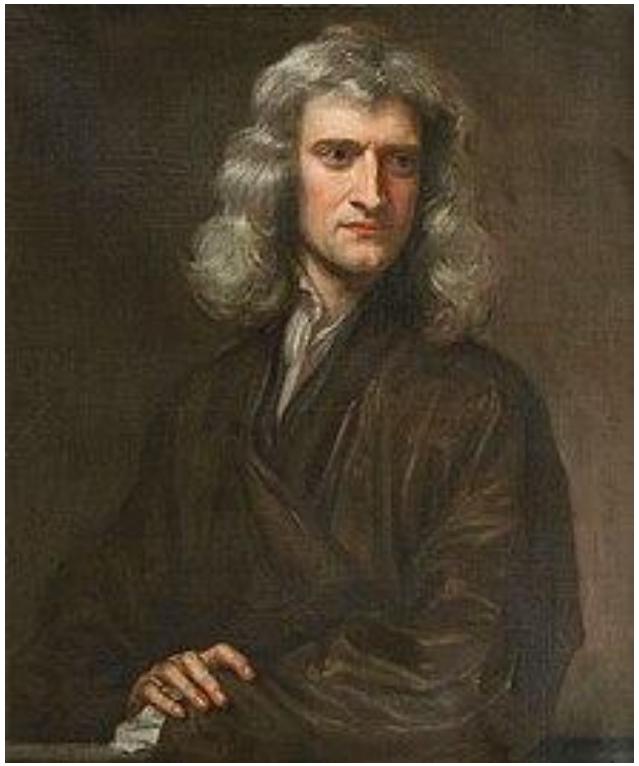
$$\boldsymbol{\mathcal{F}} = \mathbf{M}_x(\mathbf{q})\ddot{\mathbf{x}} + \mathbf{C}_x(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}_x(\mathbf{q})$$

These two are related as

$$\mathbf{Q} = \mathbf{J}^{\mathsf{T}}(\mathbf{q})\boldsymbol{\mathcal{F}}$$

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Newton-Euler formulation for dynamics

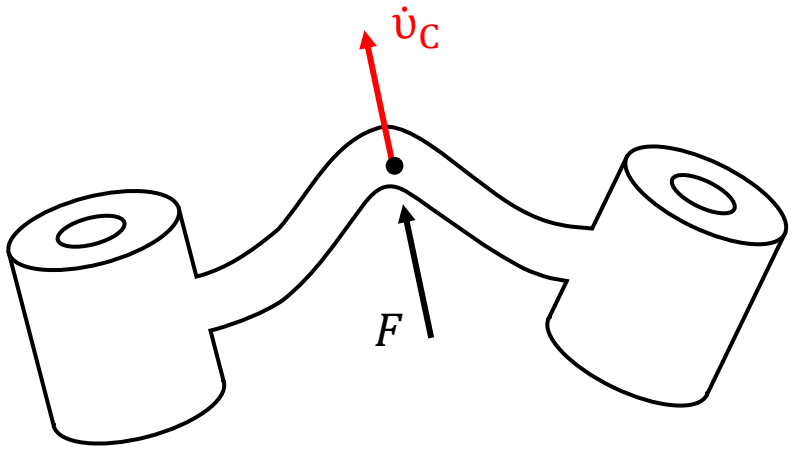




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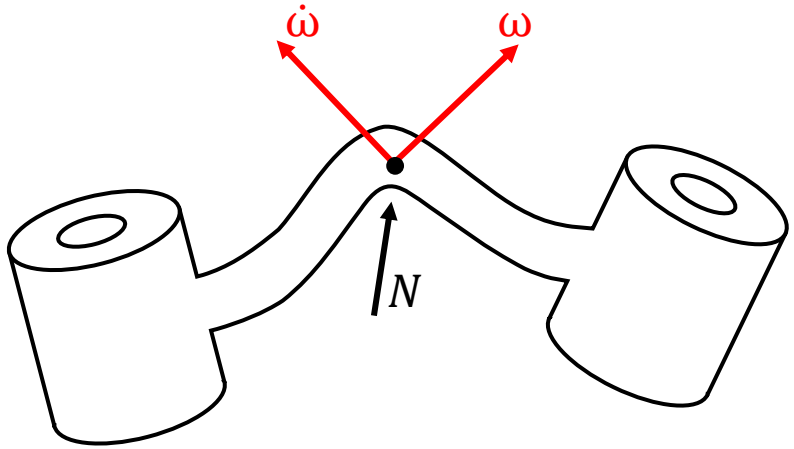
Newton's equation and Euler's equation

Consider a rigid body (such as the link of a robot) as shown below



Acceleration of the centre of mass is $\dot{\upsilon}_C$

The force that causes this acceleration is $F = m\dot{\upsilon}_C$



The body is rotating with an angular velocity of ω and acceleration of $\dot{\omega}$

The moment that causes this motion is $N = {}^{C}I\dot{\omega} + \omega \times {}^{C}I\omega$

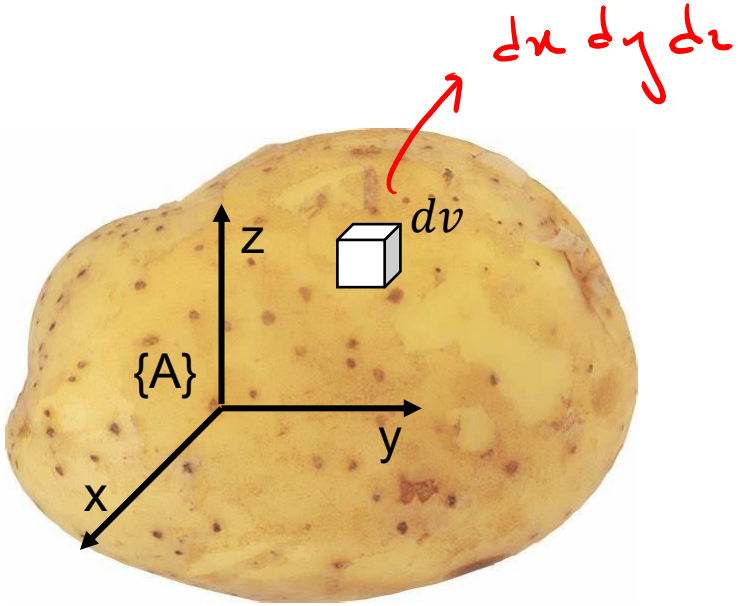
${}^{C}I \rightarrow$ inertia

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Mass distribution

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The inertia tensor



Assume uniform mass density ρ

$$ {}^{A}I = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix} $$

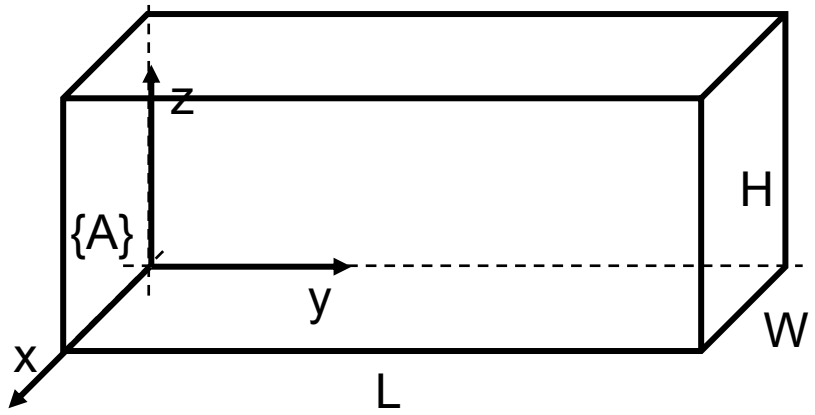
$$ I_{xx} = \iiint_V (y^2 + z^2)\rho dv \qquad I_{xy} = -\iiint_V xy\rho dv $$

$$ I_{yy} = \iiint_V (x^2 + z^2)\rho dv \qquad I_{yz} = -\iiint_V yz\rho dv $$

$$ I_{zz} = \iiint_V (x^2 + y^2)\rho dv \qquad I_{xz} = -\iiint_V xz\rho dv $$

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Example: Cuboid



Assume uniform mass density ρ

Frame {A} is at a corner

What is ${}^{A}I$?

$$I_{xx} = \iiint_V (y^2 + z^2)\rho dv = \int_0^W \int_0^L \int_0^H (y^2 + z^2)\rho \, dz \, dy \, dx$$

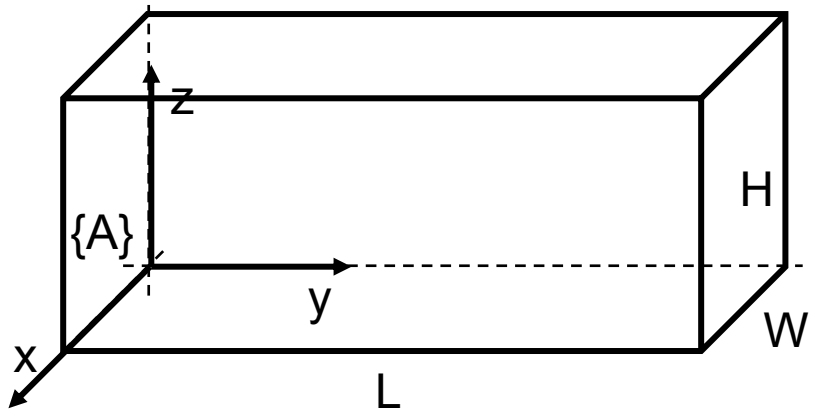
$$I_{xx} = \frac{m(L^2 + H^2)}{3}$$

$$I_{xy} = \iiint_V xy\rho dv = \int_0^W \int_0^L \int_0^H xy\rho \, dz \, dy \, dx$$

$$I_{xy} = \frac{m(WL)}{4}$$

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Example: Cuboid



Assume uniform mass density ρ

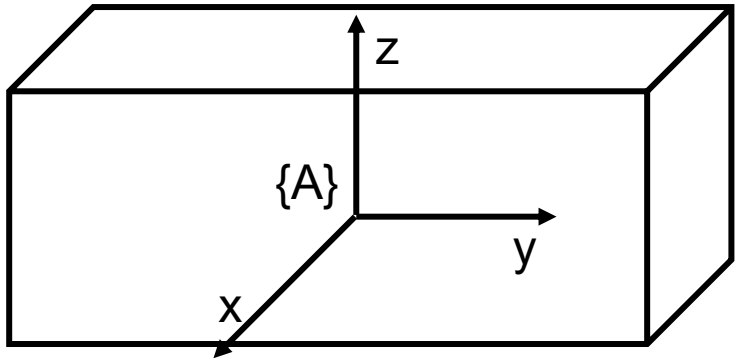
Frame {A} is at a corner

What is ${}^{\mathrm{A}}I$?

$$
{}^{\mathrm{A}}I = \begin{bmatrix} \frac{m(\mathrm{L}^2+\mathrm{H}^2)}{3} & -\frac{m(\mathrm{WL})}{4} & -\frac{m(\mathrm{WL})}{4} \\ -\frac{m(\mathrm{WL})}{4} & \frac{m(\mathrm{W}^2+\mathrm{H}^2)}{3} & -\frac{m(\mathrm{HL})}{4} \\ -\frac{m(\mathrm{HW})}{4} & -\frac{m(\mathrm{HL})}{4} & \frac{m(\mathrm{L}^2+\mathrm{W}^2)}{3} \end{bmatrix}
$$

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What if frame {A} is at the centroid?



Assume uniform mass density ρ

Frame {A} is at the centroid

What is ${}^{\mathrm{A}}I$?

$$
{}^{\mathrm{A}}I = \begin{bmatrix} \frac{m(\mathrm{L}^2 + \mathrm{H}^2)}{12} & 0 & 0 \\ 0 & \frac{m(\mathrm{W}^2 + \mathrm{H}^2)}{12} & 0 \\ 0 & 0 & \frac{m(\mathrm{L}^2 + \mathrm{W}^2)}{12} \end{bmatrix}
$$

The frame of reference is important for the inertia tensor

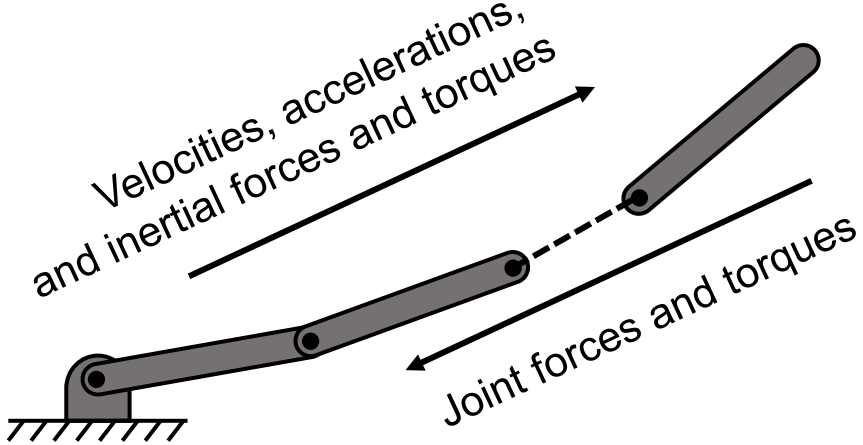
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Iterative Newton-Euler Algorithm

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Iterative Newton-Euler Algorithm

- Allows real-time computation of forward and inverse dynamics.
- The algorithm iteratively computes dynamics by progressing along links
  - Outward: compute velocity and accelerations
  - Inward: compute forces and torques



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The problem

- Given:
  - Joint positions, velocities and accelerations $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$.
  - Kinematics and mass distribution.
- Aim:
  - To calculate joint torques $\mathbf{Q}$ required to cause this motion.

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What does the outward iteration achieve?

- We start with $\mathrm{q}, \dot{\mathrm{q}}, \ddot{\mathrm{q}}$.
- We propagate velocities and accelerations through links, to obtain ${}^{i}\omega_i$, ${}^{i}\upsilon_i$, ${}^{i}\dot{\omega}_i$ and ${}^{i}\dot{\upsilon}_i$.
- Then, using our knowledge of where the centre of mass of the link is ${}^{i}\mathbf{t}_{C_i}$ we change the representation to the centre of mass frame of each link $\{\mathrm{C}_i\}$. This gives us ${}^{i}\dot{\upsilon}_{\mathrm{C}_i}$.
- Since we know the mass distribution of each link ${}^{\mathrm{C}_i}I$, we can now apply Newton's equation and Euler's equation to obtain

$$F_i = \mathrm{m}_i \dot{\upsilon}_{C_i} \quad \text{and} \quad N_i = {}^{\mathrm{C}_i}I\dot{\omega}_i + \omega_i \times {}^{\mathrm{C}_i}I\omega_i$$

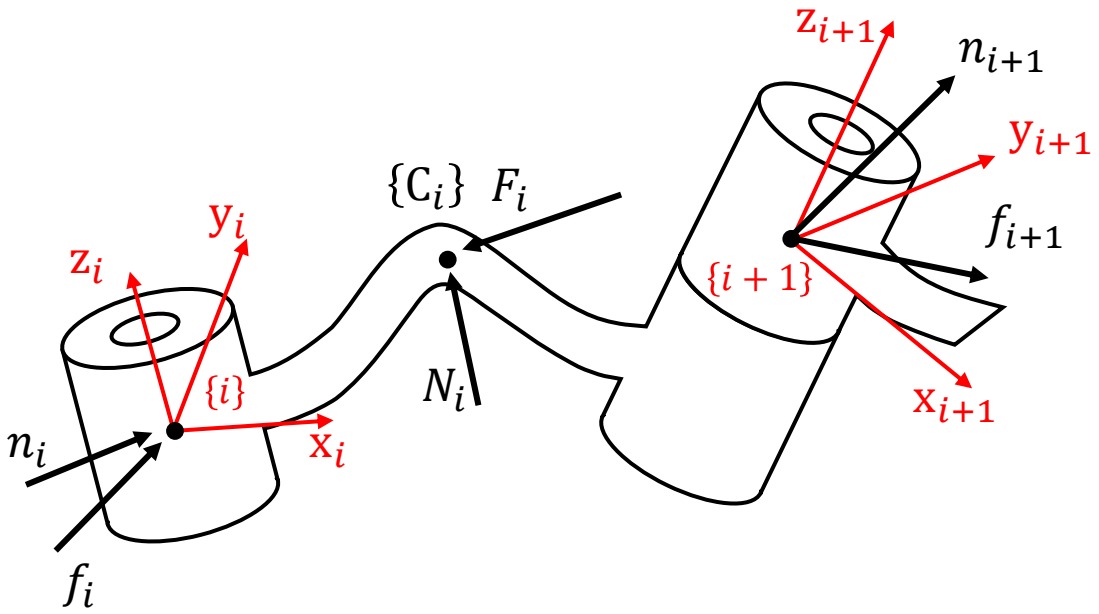
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Inward iteration

- Having computed the forces and torques on each link, we now want to compute the joint torques that result in these.
- We do this by writing force-balance and moment-balance equations.

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Balancing force and moment for a link



Balance of forces

$$^{i}F_i = {}^{i}f_i - {}^{i}\mathrm{T}_{i+1}\,{}^{i+1}f_{i+1}$$

Balance of moments

$$^{i}N_i = {}^{i}n_i - {}^{i}n_{i+1} + (-{}^{i}\mathrm{P}_{C_i}) \times {}^{i}f_i - ({}^{i}\mathrm{P}_{i+1} - {}^{i}\mathrm{P}_{C_i}) \times {}^{i}f_{i+1}$$

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Closed form solution exists in simple cases

Homework:

- Consider the two-link manipulator
- Assume that the mass of each joint is a point-mass as we did in the case of the Lagrangian formulation.
- Go through the iterations of the Newton-Euler formulation and determine joint torques.
- Verify that the expressions are the same as those that we obtained using the Lagrangian formulation.

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Computational considerations

Iterative Newton-Euler formulation

- $104n - 77$ multiplications
- $92n - 70$ additions

Lagrangian formulation

- $32n^4 + 86n^3 + 171n^2 + 53n - 128$ multiplications
- $25n^4 + 66n^3 + 129n^2 + 42n - 96$ additions

If n = 6

ne ℓ

547 vs 66394 multiplications

482 vs 51456 additions

From the handbook of industrial robotics

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What about forward dynamics?

Given: joint forces & torques $\mathbf{Q}$.
Aim: to determine the motion of the manipulator $\mathbf{q}$, $\dot{\mathbf{q}}$, $\ddot{\mathbf{q}}$.

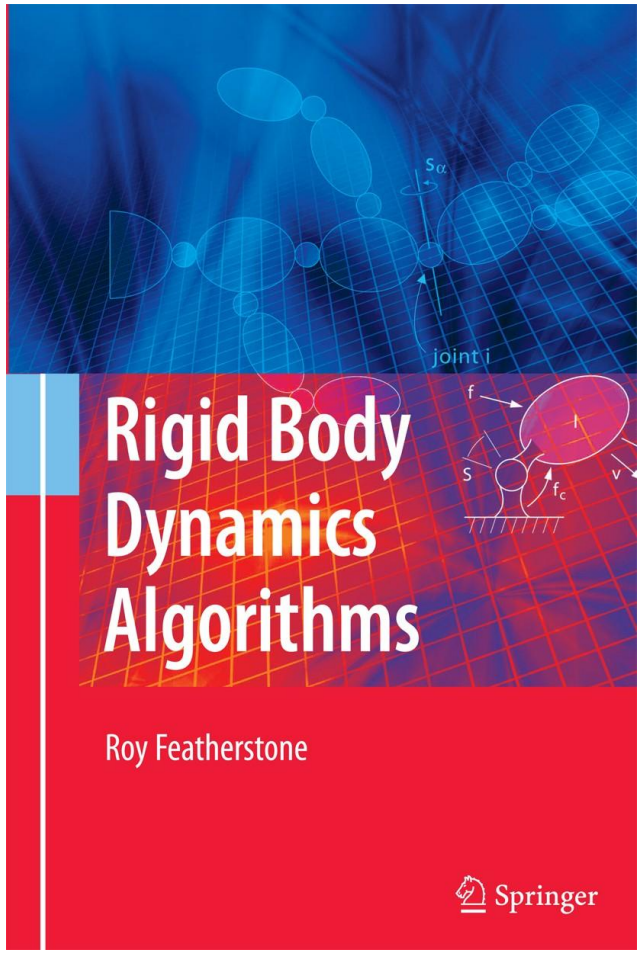
M $n \times n$

$\det(M) > 0$

$Q = M\ddot{q} + C + g + \ldots$

$$\ddot{\mathbf{q}} = \mathbf{M}(\mathbf{q})^{-1}(\mathbf{Q} - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) - \boldsymbol{g}(\mathbf{q}) - \boldsymbol{f}(\dot{\mathbf{q}}) - \mathbf{J}^{\mathrm{T}}(\mathbf{q})\boldsymbol{w})$$

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Pinocchio

Efficient and versatile rigid body dynamics algorithms

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Summary

- Why it is important to study dynamics
- Two commonly used methods to study dynamics
  - Lagrangian formulation – based on energies
  - Newton-Euler formulation – based on forces and torques
- Computational costs of each type – NE is better

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Next week

Introduction to control

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